

Problem 1: Use R and B counters; repeats are allowed. A test scores one point for each matching position. Say only the total score. Use three positions. Leave RBR visible and score the tests RRR, BRB, and RBR before hiding any secret.

Test	Score for secret RBR
RRR	
BRB	
RBR	

Problem 2: Hide a three-place secret and play. Keep every test and score visible; add rows on scrap paper if needed. Once only one secret fits, name it; naming costs no test. Reveal the secret and check your record. Swap jobs for another round.

Round 1			Round 2		
Test	Code tested	Score	Test	Code tested	Score
1			1		
2			2		
3			3		
4			4		

Problem 3: Build each visible secret below. Test RRR, BRR, and RBR against it. Compare each changed test with RRR, not with the previous row. Record all three scores.

Secret	RRR score	BRR score	RBR score
BRR			
RBR			
RRB			
BBB			

Problem 4: Hide a new secret. Test RRR, BRR, and RBR and record their scores. Use the score changes to choose R or B for positions 1 and 2. Use the RRR score to find how many Rs still belong in position 3. Name the secret, then reveal and check it.

Test	Code tested	Score
1		
2		
3		

Position 1: Position 2: Position 3: My secret:

Problem 5: Suppose RRR scores 2. Build the three possible secrets BRR, RBR, and RRB. Test each row below against all three. Circle equal scores within a row. Add two tests of your own.

Test	BRR score	RBR score	RRB score
RRR			
BRR			
BBR			
BBB			

Problem 6: Try a different first test, BRB. Make each secret below, then swap R with B at positions 1 and 3 in both the secret and the test. The changed test is RRR. Record the changed secret and check both scores.

Original secret	Score for BRB	Changed secret	Score for RRR
RRB			
BBB			
BRR			

What changed in the positions? What happened to the score?